CS23334 Fundamentals of Data Science Lab

Experiment 8

Conduct Z-Test for Given Data

Objective: To test whether the average weight of a species of birds differs from 150 grams.

Procedure:

1. **Null Hypothesis (H_0):** The average weight of the birds is 150 grams.
2. **Alternative Hypothesis (H_1):** The average weight of the birds is not 150 grams.
3. **Sample:** Measure the weights of 30 birds randomly selected from the population.
4. **Z-Test:** Conduct a Z-test to compare the sample mean to 150 grams.
5. **Decision Rule:** Use a significance level of $\alpha = 0.05$.

Code:

```
import numpy as np
import scipy.stats as stats

sample_data = np.array([
    152, 148, 151, 149, 147, 153, 150, 148, 152, 149,
    151, 150, 149, 152, 151, 148, 150, 152, 149, 150,
    148, 153, 151, 150, 149, 152, 148, 151, 150, 153
])

population_mean = 150

sample_mean = np.mean(sample_data)
sample_std = np.std(sample_data, ddof=1)

n = len(sample_data)

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic)))

print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")

alpha = 0.05
if p_value < alpha:
    print("Reject the null hypothesis: The average weight is significantly different from 150 gra
else:
    print("Fail to reject the null hypothesis: There is no significant difference in average weig
```

OUTPUT:

```
Sample Mean: 150.20
Z-Statistic: 0.6406
P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from
```

Result:

The Z-test was successfully conducted on bird weight data. The computed Z-statistic fell within the acceptance region, failing to reject the null hypothesis (H0). It was concluded that there is no significant difference in average bird weight from 150 grams at alpha = 0.05.